

Suggested Teaching Guideline for Cyber Forensics PG-DITISS August 2025

Duration: 18 Classroom hrs + 22 Lab hrs (Total: 40 hrs)

Objective: To introduce the student to Cyber Forensics and cyber laws.

Evaluation method: Theory exam – 40% weightage
Lab exam – 40% weightage
Internal Assessment – 20% weightage

List of Books / Other training material

Courseware: No specific courseware for the module, faculty may share some coursematerials.

Reference:

- Guide to Computer Forensics and Investigations Computer Forensics and Investigations/nelson
- Digital Forensic: For Mumbai University by Dr. Nilakshi, Dr. Dhananjay

Note: Every Session is having 2 Hours of theory and each Lab is having 2 Hours of duration, unless indicated Otherwise.

Conduct lab sessions based on case studied. Faculty can ask students to get some casestudies and choose among that for explanation.

Session 1: Introduction

- Overview - Computer Forensics
- Difference – Computer Crime & Un-authorized activities
- Cyber Laws
- Process of Computer Forensics (six)
- Need for forensics investigator

Session 2:

- Computer Forensics Involves
 - Preservation
 - Identification
 - Extraction
 - Documentation
 - Interpretation

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- Goals of Forensics Analysis
- Types of Cyber Forensics Techniques
- Cyber forensics Procedures
- Preparation
 - What to do before the incident
 - Incident response plan
 - Incident response team
 - Detecting Incidents
 - Chain of custody

Lab 1 & 2:

- Simulating Incident Response and Chain of Custody Handling

Session 3:

- Evidence Checkout Log
- Handling Evidence
- First Response
- Formulate/Execute Response Strategy
- Forensic duplication
- Authenticate the Evidence
- Investigation
- Common Mistakes
- Detection

Lab 3 & 4:

- Demo and lab sessions on Evidence Checkout Log, Investigation and Detection.

Session 4:

- The Initial Assessment
- Incident Notification Checklist
- Hexadecimal notation
- Practical Bits
- Slight diversion
- What is use of Hexadecimal

Session 5:

- Encoding And Encryption
- The Hex Editor
- Files
- Hashing

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- Hashing DLs
- MD5 Hash collisions
- Hash Collisions
- Bit Rot

Lab 5:

- Demo and lab sessions on Encoding and Encryption.

Session 6:

- Standard Operating Procedures
- Processing Crime and Incident Scenes – Working with Windows & DOS Systems Current

Lab 6:

- Processing a Crime Scene on Windows and DOS Systems

Session 7:

- Forensics Implications
- Accreditation Standards
- Performing a Cyber Forensics Investigation
- Privacy and Cyber Forensics

Lab 7:

- Conducting a Cyber Forensics Investigation and Addressing Privacy Considerations

Session 8:

- Computer Forensics Tools:
 - Sysinternals Suite
 - FTK Forensics Tool kit
 - FTK Imager
 - OSF
 - Hex

Lab 8 & 9:

- Demo and lab sessions on Cyber Check Suit

Session 9:

- Concept of collecting and analyzing artefacts from systems that are in the active state
- Concept of collection and analysis of artefacts from Linux systems
- An introduction to Mobile forensics

Lab 10 & 11:

- Demo and lab sessions on Live system forensics & Linux Forensics